Reference Standard (440 Hz tuning)

A calibration standard is an absolute "must" for the FORMANT—especially since the previous lack of such a circuit was also criticized in a review of the FORMANT published in a well-known German music trade magazine.

In a simple manner, this "electronic tuning fork" generates the standard concert pitch "A" (440.0 Hz). Thanks to the FORMANT's diverse control options, it serves as an (almost) indispensable Due to the manifold control options, the tuning voltage of the VCOs—for instance—changes with every new program combination. The calibration standard, set to 440 Hz, ensures rapid adjustment of the keyboard.

Unless one is dealing with studio-grade equipment, the effort required for quartz standards is too costly. For small to medium-sized music synthesizers—such as those used by amateurs—a stable (conventional) sine-wave oscillator is entirely sufficient.

The Circuit

The frequency of the oscillator, whose complete circuit diagram is shown in Figure 1, is determined by a simple Wien bridge. C1, C2, B2, P2, and P2 are integrated as frequency-determining components in the positive feedback path of op-amps IC1 and IC2. P1 is connected in the negative feedback path, which provides stabilization. Op-amp 1C2 provides an adjustment of the signal level to FORMANT standards and for low-impedance decoupling. With S1, the sine wave oscillator can be "switched off."

Construction

Construction based on the layout and component placement plan shown in Figure 2 should present no difficulties whatsoever. Since the circuit—which is not particularly large—fits comfortably onto a standard Eurocard, the remaining free space can be utilized to accommodate a +9 V power supply; this can be used, for example, to enable various effects units for musicians to operate independently of batteries. The schematic and component placement plan for this power supply can be found in Appendix D. It is advisable to route the fine-tuning potentiometer P3 to the front panel so that subsequent adjustments can be made without having to remove the rear cover or extract the module. A corresponding mounting hole is provided on the front panel (Figure 3). 70-turn helical potentiometers are commercially available and are, of course, well-suited for this purpose. Their only drawback is likely their cost. The sketch in Figure 4 illustrates how this issue can be resolved more simply—that is, more economically. A Cermet helical trimmer is mounted on a perforated circuit board, which is in turn secured to the rear side of the front panel using an aluminum angle bracket. The trimmer's connections are routed via solder pins. The only critical component is R7. Should a 220-ohm PTC resistor not be available, a miniature incandescent lamp (6 V, 30 mA) may be used instead.

Calibration

P1 must be adjusted so that the amplitude of the output voltage lies within a range of 2.5 V. When measuring with a multimeter (preferably one with high input impedance), it should be noted that the measured RMS value must be multiplied by the factor $2\sqrt{2}$ (= 2.828).

P2 is used for coarse adjustment, and P3 for fine adjustment of the oscillator frequency. Due to operating temperature considerations, calibration should not be started until the unit has been running for approximately 10 minutes. Precise adjustment of P2 and P3 can be easily achieved by tuning the oscillator against a reference signal to "zero beat." An AF function generator, an electronic organ, or a tuning fork may be used as a reference signal.

More accurate results can be obtained by calibrating with a digital frequency counter, although this process is correspondingly time-consuming given a gate time of 5 to 10 seconds. With a view to its use in conjunction with other (mechanical) musical instruments, a calibration frequency of 440 Hz was selected. This corresponds to the "Kammerton" (concert pitch) A4. However, the sine oscillator oscillates reliably within a range of 250 Hz to 750 Hz, allowing it to be tuned to a frequency other than 440 Hz if desired.

Wiring

The wiring connections between the circuit board and the front panel require no further explanation.

However, a few remarks regarding the "placement" of the calibration standard within the internal wiring of the FORMANT music synthesizer are in order:

The internal output (IOS) of the calibration standard module is best wired to the IS input—an input that is also connected to one or more outputs of the VCAs. Via the EOS output on the front panel, the tuning tone is also available—for instance, for an additional "VCO Bank" housed in a separate enclosure.

Application in the FORMANT

The tuning procedure for the FORMANT VCOs is described in detail in the second section of the first FORMANT manual.

To tune the system, set the VCFs and VCAs to full transparency (open) with maximum gain. Next, while holding down the "A" key on the keyboard, adjust the individual FINE and COARSE controls to achieve a zero beat. With this done, the FORMANT is ready to be played alongside other musical instruments. When tuning to chords, simply count down—starting from the "A" reference—the number of semitones by which the target VCO is intended to sound higher, and then adjust that VCO to a zero beat. For example, to tune to a major third interval, set VCO1 to "A" and VCO2 to "7"—that is, four semitones lower—and adjust both to a zero beat. VCO2 will then sound exactly four semitones higher than VCO1. This eliminates the need to "memorize" the individual tuning settings for each VCO, thereby greatly simplifying the tuning process. Even after the initial adjustment, the resulting chord is usually so pure that further fine-tuning is rarely necessary.

Calibration Standard Bill of Materials

Resistors (Carbon Film, 5%):

- R1 = 560 Ω
- R2 = 10 kΩ
- R3, R4 = 100 kΩ
- R5 = 56 kΩ
- R6 = 470 Ω
- R7 = PTC 220 Ω (see text)

Potentiometers:

- P1 = 1 kΩ (Trimmer)
- P2 = 10 kΩ (Trimmer)
- P3 = 470 Ω (see text)

Capacitors:

- C1, C2 = 47 nF (MKH/MKS)
- C3, C4 = 470 pF (MKH/MKS)

Semiconductors:

- IC1...IC3 = 741 (Mini-DIP)

Miscellaneous:

- S1 = Miniature Toggle Switch (SPDT)
- 1 x 3.5mm Jack Socket
- 1 x 31-pin Header Strip or Solder Pins

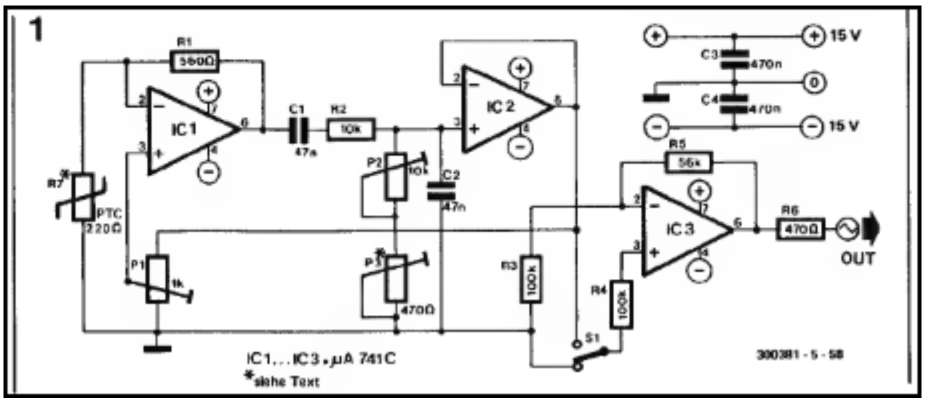
Figures

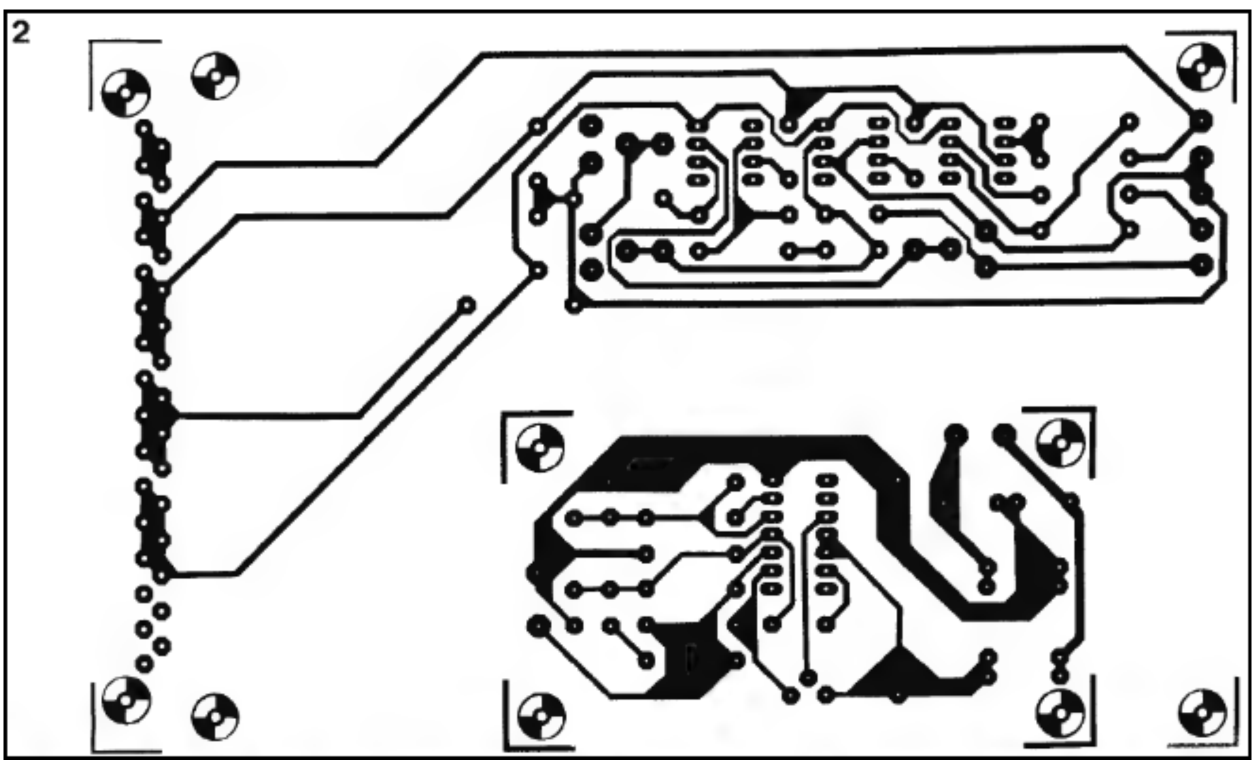
- **Figure 1.** Calibration Standard for the FORMANT. It consists of a sine-wave oscillator tuned to 440 Hz (concert pitch "A").
- **Figure 2.** Layout and component placement diagram for the calibration standard. Since this circuit board is also designed in the Eurocard format, sufficient space remains for a +9 V power supply. The circuit diagram, PCB layout, and parts list are provided in Appendix D.

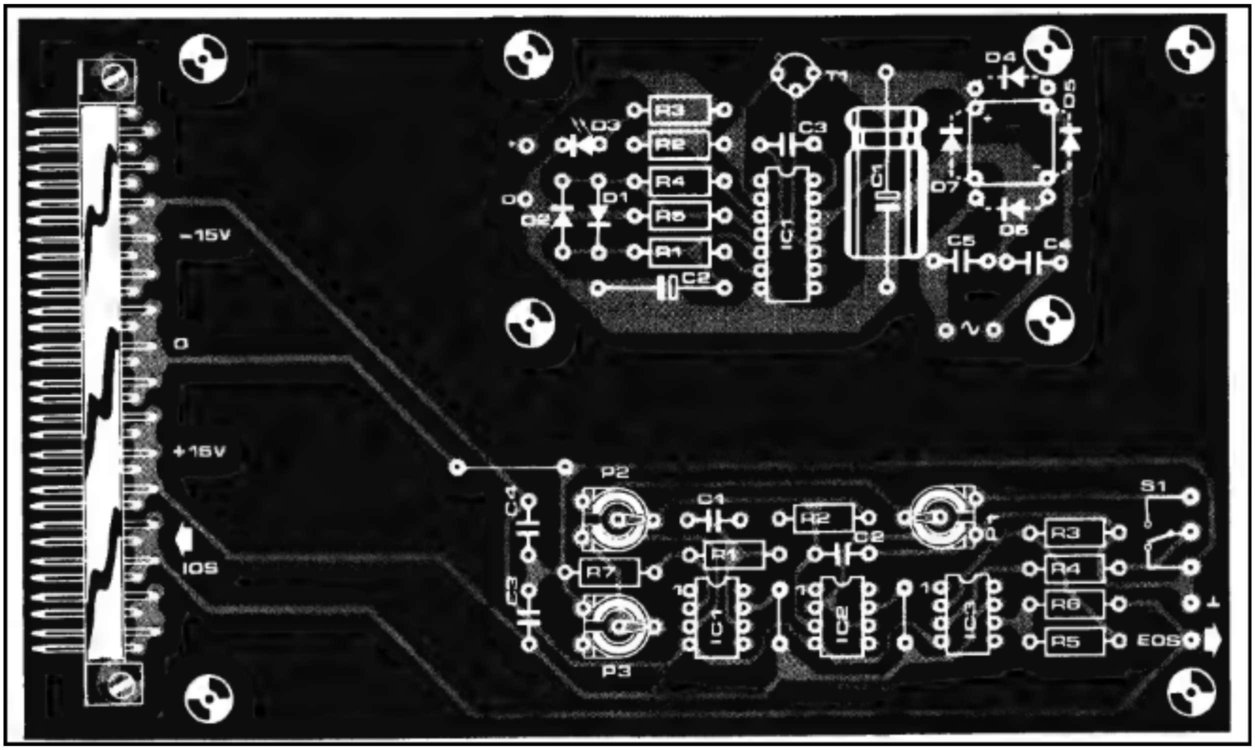
- **Figure 3.** Proposed front panel layout (80% of original size) for the calibration standard.
- **Figure 4.** Proposed mounting arrangement for a cermet helical trimmer potentiometer, in cases where a 10-turn helical potentiometer is not a viable option.

References:

- C. Chapman: FORMANT Music Synthesizer, Elektor Verlag
- G. Dellmann: FORMANT SYNTHESIZER, Fachblatt – Music Magazine (Section: Music – TÜV), No. 53, August 1977
- H. Tünker: Electronic Pianos and Synthesizers, Franzis Verlag, Munich, RPB Series No. 302 (1975, 1st ed.) pp. 109, 110

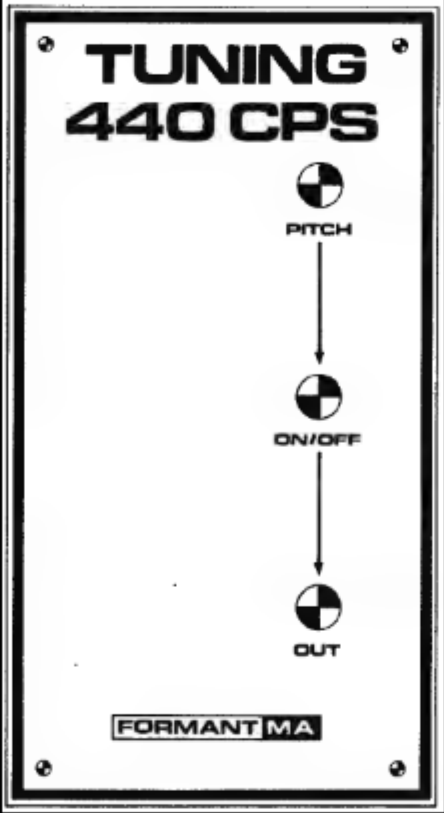




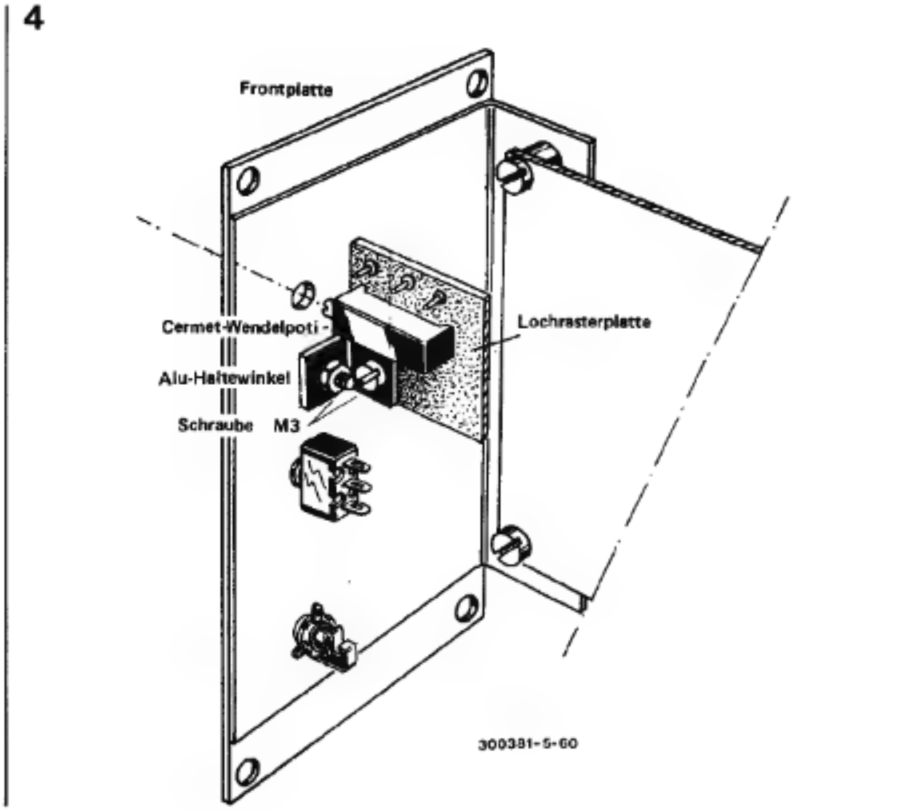


-15V
0
+15V
IOS
D3
D2
D1
R3
R2
R4
R6
R1
C2
C3
IC1
T1
C1
D4
D5
D7
D6
C5
C4
P2
C4
C1
R2
C2
R1
R7
C3
P3
IC1
IC2
IC3
S1
R3
R4
R6
R5
EOS

3


TUNING
440 CPS
PITCH
ON/OFF
OUT
FORMANT MA

4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Cermet-Wendelpoti
Lochrasterplatte
Alu-Haltewinkel
Schraube M3
300381-5-60

A Brief Word

Fundamental information regarding the scope of the FORMANT's modules has already been covered in the first FORMANT book, as well as in the introduction to this volume. However, since this topic presents a rather significant challenge for many synthesizer users, a final attempt to provide an answer is certainly warranted.

Very few synthesizer enthusiasts know exactly—right from the start—which modules they will need, or in what quantities. As a rule, those encountering a complex electronic musical instrument for the first time typically lack concrete ideas regarding the appropriate scope of their system. Fundamentally, the overall scope of a synthesizer system is determined by its intended application. An audio/film hobbyist who merely wishes to provide an acoustic backdrop for their slide and film shows requires a different module configuration than an (electronic) jazz musician. When building a system oneself, the amount of time one is able to dedicate to the hobby also plays a significant role. Last but not least, an individual's financial situation is also a crucial factor when configuring a system. However, much can be done to address this through meticulous planning (e.g., by creating a comprehensive parts list for the entire system). After all, when purchasing—for instance—100 resistors, the unit cost per resistor is lower than when purchasing only 10.

One important point should not be overlooked during the planning phase: the potential for future expansion of the initial system. The desire to expand the system inevitably arises once all its musical capabilities have been fully explored. Therefore, this factor must be given careful consideration during the mechanical design phase and when selecting a housing. The modular architecture of the FORMANT is ideally suited for—and practically invites—construction in stages. This approach offers a further advantage that may not be immediately obvious: by starting with a manageable scope, one can achieve musically meaningful results much more quickly, while simultaneously growing into the system as it is gradually expanded later on.

So, exactly how many of each module does one actually need? Experience shows that one can never have enough VCOs, VCAs, and ADSRs. The exact number of VCFs required is difficult to determine, as it depends—particularly—on the intended application; this applies equally to several other modules described in this book. However, it certainly does no harm to build at least one of each of these specialized circuits (ring modulator, envelope follower, sample & hold, etc.) in order to discover the musical potential inherent in them. Only once you have gained a clear understanding of this can the specific number of new modules be decided upon. The accompanying table can serve as a guideline during this decision-making process. It is based on empirical data derived from typical synthesizer systems. The "basic configuration" corresponds roughly to the circuits described in Book 1. The setup designated in the table as a "medium system" already falls into the high-end category when compared to commercial equipment. This level of sophistication offers ample scope for both the amateur enthusiast and the adventurous player. The "large system" closely mirrors the configurations found in professional recording studio synthesizers. Finally, when assembling your own system, you might also choose to use the equipment setups of well-known synthesizer players as a point of reference.

In any case, amidst all these considerations, one objective must never be lost sight of: a synthesizer exists for the sole purpose of making music. It serves as an effective tool for providing beginners with an entry point into the world of musical acoustics—and, by extension, into the realm of electronic music. The true art lies in translating one's own imagination into concrete musical forms. Keeping records of interesting sounds helps in building a personal "idea bank." This can be done either in writing—using the methods outlined in Appendix A—or by using a cassette recorder as an "active memory."

One final thought: despite all the modernization and expansion of individual modules, it is ultimately the player themselves who shapes physical vibrations into music. New modules, in and of themselves, do not automatically foster greater creativity. Experience has already proven this to be true: the more module cabinets and keyboards that cluttered the stage, the more the imagination tended to wither away. Being a gear fetishist is no guarantee of a "creative sound." On the contrary: with just two or three FORMANT modules (e.g., a VCF and a Noise module), a microphone, and an acoustic instrument, it is already possible to venture into new acoustic dimensions. Jiří Klásek once aptly described this potential when he said: "Through electronics to more music—and not the other way around."

	Module Title	Basic Fit-out	Intermediate System	Large system
FORMANT - Book 1 + 2	VCO	3	4…6	7…12
	VCF (12dB)	1	2	3…4
	VCF (24dB)	1	2	4…5
	Dual-VCA	1	2…3	5…10
	ADSR	1…2	3…6	8…15
	LFOs, VC-LFOs, LF-VCO	1	2	3…4
	NOISE or DNG	1	2	4
	RFM	0…1	1…2	2…4
	COM	1	2	2…4
FORMANT - Book 1 + 2	Controllers (analog or digital	1	2	3…6

	Keyboard, Joystick, Touch Plate, Ribbon Controller, etc.)			
FORMANT - Extensions (Book 2)	Stroke Control	0	0…1	1…2
	NPD	0…1	1…2	2..4
	Ring Modulator	0…1	1…2	4
	Phase Shifter	0	1…2	3…5
	Krimisizer	0…1	0…2	0…3
	(Digital) Reverb	0	0…1	1…3
	KOV/KB-Gate	0…1	1…2	2
	Multiple Jacks	0…1	1…2	3…6
	ADSR-Controller	0	1…2	3…8
	ENV-F	0	1	2
	Waveform Processor	0	1	2
	S & H	0…1	2	4
	Mixer	0	1…2	3…6
	440 Cps Tuning	0	1	1…2
	(DIgital) Sequencer	0…1	1…2	2…5